NCERT Summary: Topic 12: Natural Hazards & Disasters — Raw Notes

Source Integration

- NCERT Class 11 — India: Physical Environment, Chapter 6: Natural Hazards and Disasters: Causes, Consequences and Management

Core Factual & Conceptual Architecture

1. Hazard versus Disaster & Classification

- Natural Hazard: An element or circumstance in the natural environment with the potential to cause harm, whether permanent or swift.
- Natural Disaster: A relatively sudden event causing large-scale, widespread death, property loss, and social disruption beyond human control.
- Human Contribution: Disasters are often intensified or directly caused by human activity, including deforestation, unscientific land use, pollution, and greenhouse gas emissions.
- Four Natural Disaster Categories:
  - Atmospheric: Blizzards, tropical cyclones, droughts, heat waves, and tornadoes.
  - Terrestrial: Earthquakes, volcanic eruptions, landslides, and avalanches.
  - Aquatic: Floods, storm surges, and tsunamis.
  - Biological: Insect infestations, locust swarms, and viral or bacterial epidemics.

2. Earthquakes & Tsunami Mechanics

- Earthquake Genesis: Tectonic-origin earthquakes resulting from sudden energy release along fault lines are the most devastating. India is divided into five seismic damage risk zones, with the highest risk along the Himalayan arc and Northeast states.
- Tsunami Dynamics: Seismic sea waves caused by abrupt sea-floor displacement. In deep water, wavelengths are long and heights are low, but upon entering shallow coastal waters, wavelengths shorten and wave heights surge up to 15 meters or more.

3. Tropical Cyclones & Floods

- Tropical Cyclone Formation: Intense low-pressure systems between 30 degrees north and 30 degrees south functioning as heat engines, driven by latent heat release from warm ocean moisture and stabilized by the Coriolis force.
- Flood Mechanics: Inundation caused by runoff exceeding river channel capacity, storm surges, or prolonged rainfall. High flood-prone states in India include Assam, Bihar, and West Bengal, covering roughly 40 million hectares.

4. Drought Typologies & Severity in India

- Four Drought Types: Meteorological (rainfall deficit), agricultural (low soil moisture), hydrological (aquifer depletion), and ecological (ecosystem productivity failure).
- Severity Zones: Extreme drought in western Rajasthan and Kachchh; severe drought in eastern Rajasthan, Madhya Pradesh, and interior Andhra Pradesh; moderate drought across surrounding semi-arid tracts.

5. Landslides & Disaster Management Framework

- Landslide Vulnerability: Highly localized hazards concentrated in the unstable young Himalayas, Andaman and Nicobar, and steep slopes of the Western Ghats.
- Disaster Management Phases: Pre-disaster preparedness and mapping, during-disaster rescue and relief, and post-disaster rehabilitation. Governed in India by the Disaster Management Act of 2005.

✍ Notes & Updates